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IMPACT OF PRICING STRATEGY ON SALES PERFORMANCE OF PHARMACEUTICAL FIRM IN NIGERIA

OYEGBAMI O. G.; OLADEJO M. O.; & ONIGBINDE I. O.

Department of Marketing, Ladoke Akintola University of Technology, Ogbomoso, Oyo State, Nigeria.

Corresponding Author:
mooladejo@lautech.edu.ng

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Abstract

This study investigates the impact of pricing strategies on the sales performance of pharmaceutical firms in Nigeria. The study specifically examined the effects of cost-plus pricing, value-based pricing, competitive pricing, penetration pricing, and skimming pricing on sales performance. Using a qualitative research design, primary data were collected through structured questionnaires administered to management and marketing staff of selected pharmaceutical firms. The data were analyzed using multiple regression analysis to determine the impact of pricing strategy on sales performance of pharmaceutical firm in Nigeria. The results revealed that pricing strategy has a significant influence on sales performance, with an R value of 0.849 and an R^2 of 0.721, indicating that approximately 72% of the variation in sales performance is explained by pricing strategy variables.

Introduction

The pharmaceutical industry in Nigeria plays a crucial role in ensuring access to essential medicines and promoting public health. However, the sector continues to face challenges such as rising production costs, inflation, exchange rate volatility, and supply chain disruptions, all of which significantly affect pricing decisions (Uzor, Okoye, & Okafor, 2024). These factors have caused medicine prices to surge across both public and private sectors, thereby influencing consumer purchasing patterns and firm profitability. As observed by Danraka, Yusuf, and Bello (2022), medicine affordability in Nigeria has declined due to cost inflation and currency depreciation, which calls for strategic pricing decisions by pharmaceutical firms to sustain sales growth and market share.

Among the predictors, cost-plus pricing exhibited a strong and statistically significant positive effect on sales performance (B = 0.861, t = 10.491, p < 0.001), implying that firms adopting cost-based approaches tend to achieve higher sales growth and profitability. Conversely, competitive pricing showed a significant negative effect (B = -0.231, t = -2.227, p = 0.029), suggesting that excessive price competition may reduce profitability and long-term market stability. Penetration pricing had a marginally negative effect (B = -0.158, t = -1.960, p = 0.054), while value-based pricing (B = 0.118, p = 0.171) and skimming pricing (B = 0.072, p = 0.364) had positive but insignificant effects on sales performance. The findings indicate that cost-plus pricing is the most effective strategy for enhancing sales performance in Nigeria's pharmaceutical industry, while aggressive price competition and penetration tactics may hinder sustainable profitability. The study concludes that firms should adopt cost-reflective and value-oriented pricing policies to balance profitability with consumer affordability. It also recommends that managers employ data-driven pricing decisions to adapt to market changes and strengthen sales performance in a highly competitive and regulated environment.

Keywords: Pricing strategy, cost-plus pricing, competitive pricing, value-based pricing, sales performance, pharmaceutical firms, Nigeria.

Pricing strategy refers to the deliberate process of setting prices that align with organizational objectives, market conditions, and consumer perceptions (Adegoke & Ajibola, 2022). In the pharmaceutical sector, pricing is shaped not only by cost recovery but also by ethical considerations, government regulations, and competitive forces. For example, Okafor (2020) emphasized that pharmaceutical firms in Nigeria often adopt cost-plus or value-based pricing to reflect production costs while ensuring drugs remain accessible to patients. Similarly, Bello and Oyetunde (2023) found that competitive and penetration pricing approaches helped firms attract a larger market share in the face of rising generic drug competition. Therefore, the choice of pricing strategy becomes a key determinant of how pharmaceutical firms achieve profitability and maintain consumer trust.

Sales performance measures how effectively a company's pricing and marketing efforts translate into financial outcomes such as revenue growth, sales volume, and market share (Oyelade & Ogunleye, 2023). In Nigeria's pharmaceutical market, sales performance depends heavily on price affordability and perceived product value. According to Usman and Bello (2021), when medicine prices exceed consumers' purchasing power, sales volume drops even for essential drugs, especially among low-income earners. However, value-based and competitive pricing strategies have been shown to enhance customer retention and boost revenue by improving

perceived fairness and quality. Hence, pricing decisions are not merely financial choices but strategic levers that shape long-term customer loyalty and firm success.

Statement of the Problem

Despite the crucial role of pricing, limited empirical studies have focused specifically on how pricing strategies influence the sales performance of pharmaceutical firms in Nigeria. Most prior research has addressed affordability and access rather than firm-level performance outcomes (Aliyu & Ahmed, 2019; Okafor, 2020). As noted by Adegoke and Ogundele (2022), Nigerian pharmaceutical firms need to adopt flexible pricing models to navigate inflationary pressures while maintaining affordability. This knowledge gap highlights the need for a comprehensive investigation into how different pricing approaches—such as cost-plus, competitive, and value-based pricing—affect sales growth and profitability within the Nigerian context.

Therefore, this study seeks to examine the impact of price strategy on sales performance of pharmaceutical firms in Nigeria. Specifically, it will assess how various pricing strategies influence key performance indicators such as sales growth, customer retention, and market share. The findings of this study guides pharmaceutical managers in formulating optimal pricing policies and assist policymakers in balancing profitability with consumer welfare. Ultimately, the research contributes to understanding how strategic pricing decisions can drive sustainable performance in Nigeria's competitive and economically volatile pharmaceutical industry.

Objective of the Study

The study examined the impact of pricing strategy on sales performance of pharmaceutical firm in Nigeria

Review of Previous Studies

Several scholars have examined the relationship between pricing strategy and sales performance across different sectors, including pharmaceuticals, manufacturing, and fast-moving consumer goods (FMCGs). Their findings reveal that the type of pricing strategy adopted—whether cost-plus, value-based, or competitive—has a direct effect on firms' market positioning, profitability, and customer retention.

In a recent study, Oyelade and Ogunleye (2023) investigated the effect of various pricing strategies on the sales performance of Nigerian pharmaceutical companies. Using survey data from 150 marketing and sales executives across selected firms in Lagos and Ogun States, the study employed regression analysis to determine the relationship between cost-plus, competitive, and value-based pricing strategies and sales outcomes. The results indicated that cost-plus and value-based pricing

significantly improved revenue growth and customer retention. The authors concluded that firms that align pricing strategies with perceived customer value tend to achieve stronger sales performance than those relying solely on cost recovery approaches.

Similarly, Adegoke and Ajibola (2022) examined the influence of pricing decisions on market competitiveness in Nigeria's drug industry. Their findings revealed that firms that adopted competitive pricing strategies—where prices are adjusted in response to rival firms' pricing behavior—recorded better market share and customer loyalty. However, excessive price reductions, often used as a penetration strategy, negatively impacted profitability in the long run. This aligns with the conclusion of Bello and Oyetunde (2023), who reported that competitive pricing enhanced short-term sales performance but could be detrimental to long-term financial sustainability if not properly managed.

From a regional perspective, Usman and Bello (2021) conducted an econometric study on the effect of pricing on pharmaceutical product demand in Nigeria. Using time series data from ten major pharmaceutical firms, their study found that value-based pricing positively influenced demand and customer trust, while cost-plus pricing had minimal effect on market share. The authors emphasized that consumers' perception of fairness and product quality plays a vital role in sustaining sales growth in the pharmaceutical industry.

In another study, Okafor (2020) analyzed the role of price regulation and government control on the sales performance of pharmaceutical firms in Nigeria. The study employed a mixed-method approach and found that government-imposed price caps on essential drugs limited profitability but encouraged broader access to healthcare. Okafor argued that firms must innovate their pricing models to comply with regulation while maintaining competitiveness, perhaps by differentiating between essential and non-essential drug pricing strategies.

International evidence also supports the link between pricing strategy and firm performance. For instance, Kumar and Sharma (2021) examined the impact of pricing strategies on the sales performance of pharmaceutical firms in India. Their results showed that value-based and differential pricing contributed positively to market expansion and revenue growth, while cost-based pricing was less effective in retaining customers in competitive markets. Similarly, Aliyu and Ahmed (2019) found that penetration and psychological pricing strategies enhanced brand loyalty in the Nigerian pharmaceutical industry, particularly among generic drug producers.

More recently, Uzor, Okoye, and Okafor (2024) highlighted how inflation and currency depreciation have increased medicine prices in Nigeria, reducing affordability and affecting sales volume. Their study pointed out that while pharmaceutical firms attempt to pass increased costs onto consumers through cost-plus pricing, this often leads to reduced product demand. The authors

recommended the adoption of value-based and differentiated pricing mechanisms that consider both consumer purchasing power and competitive pressures in the market.

Previous studies consistently suggest that the pricing strategy adopted by pharmaceutical firms has a significant impact on sales performance. However, most of these studies emphasize either cost or competition, while limited attention has been given to how value perception and regulatory influences interact with pricing to affect sales outcomes. This gap highlights the need for further empirical investigation into how different pricing strategies affect key sales performance indicators such as revenue growth, market share, and customer retention in the Nigerian pharmaceutical sector.

Methodology

This study adopted a quantitative research design to examine the impact of pricing strategies on the sales performance of pharmaceutical firms in Nigeria. Primary data were collected through structured questionnaires distributed to management, sales, and marketing personnel of selected pharmaceutical companies within Lagos and Ogun States. A total of 100 questionnaires were administered, out of which 82 were duly completed and analyzed. The instrument measured variables such as cost-plus pricing, value-based pricing, competitive pricing, penetration pricing, skimming pricing, and sales performance using a five-point Likert scale. The reliability of the instrument was confirmed using Cronbach's alpha coefficient, while validity was ensured through expert review. Data were analyzed using descriptive statistics and multiple regression analysis to determine the effect of pricing strategies on sales performance, with the aid of the Statistical Package for Social Sciences (SPSS, version 26).

Result of multiple regression analysis on the impact of pricing strategy on sales performance of pharmaceutical firm in Nigeria

The multiple regression analysis in Table 1a, 1b, and 1c aimed to determine how different pricing strategies—namely cost-plus pricing, value-based pricing, competitive pricing, penetration pricing, and skimming pricing—influence the sales performance of pharmaceutical firms in Nigeria. The model summary shows a correlation coefficient ($R = 0.849$), indicating a strong positive relationship between the independent variables (pricing strategies) and the dependent variable (sales performance). The R^2 value of 0.721 suggests that 72.1% of the variation in sales performance can be explained by the combined effects of the five pricing strategies. This high explanatory power demonstrates that pricing strategy is a major determinant of sales performance in the Nigerian pharmaceutical sector. The Adjusted R^2 of 0.703 further confirms the robustness of the model, accounting for

sample size and number of predictors. This means the model is both statistically sound and practically meaningful in explaining sales variation due to pricing policies.

The ANOVA (F-test) results strengthen this conclusion, showing that the model is statistically significant at the 1% level (F = 39.335, Sig. = 0.000). This indicates that, collectively, the pricing strategies included in the regression equation significantly predict sales performance. In simpler terms, changes in pricing approach have a measurable and significant impact on the sales outcome of pharmaceutical firms. The small residual sum of squares (44.598) compared to the regression sum of squares (115.414) further confirms that most of the variation in sales performance is captured by the model. Therefore, pricing strategies are not just operational decisions but strategic levers that significantly influence firm performance in the competitive and regulated pharmaceutical industry.

The coefficients table provides deeper insight into the individual contributions of each pricing strategy. Cost-Plus Pricing has a strong positive and statistically significant effect ($\beta = 0.861$, $t = 10.491$, $p = 0.000$), suggesting that firms relying on cost-based mark-up pricing experience higher sales performance. This aligns with the operational realities of the Nigerian pharmaceutical sector, where production and importation costs are key determinants of pricing decisions. On the other hand, Value-Based Pricing has a positive but statistically insignificant effect ($\beta = 0.118$, $p = 0.171$). This implies that while pricing based on customer-perceived value could enhance sales, its impact is not yet strong enough to produce statistically significant changes in performance—possibly due to limited customer awareness or weak brand differentiation among pharmaceutical products.

The findings also reveal that Competitive Pricing and Penetration Pricing have negative coefficients, indicating that they may reduce sales performance. Competitive Pricing ($\beta = -0.231$, $p = 0.029$) shows a significant negative effect, implying that when firms aggressively lower prices to match competitors, profitability and long-term performance may decline. This may be attributed to price wars and reduced margins that often characterize the pharmaceutical industry when companies compete mainly on price rather than product quality or value. Similarly, Penetration Pricing ($\beta = -0.158$, $p = 0.054$) exhibits a marginally significant negative effect. While initially intended to attract customers with lower prices, this strategy might harm long-term sales performance by reducing perceived product quality and limiting future pricing flexibility. Skimming Pricing, although positive ($\beta = 0.072$), was statistically insignificant ($p = 0.364$), indicating that charging higher initial prices has not yielded significant sales benefits, likely due to Nigeria's price-sensitive consumer base.

The regression results reveal that pricing strategies significantly affect the sales performance of pharmaceutical firms in Nigeria, but the nature and direction of these effects differ across strategy types. The dominance of cost-plus pricing

underscores the importance of covering production costs and maintaining consistent profit margins in a sector heavily influenced by exchange rate fluctuations, import costs, and regulatory pricing. The negative impact of competitive and penetration pricing strategies highlights the dangers of price-driven competition in a market where consumers are sensitive to both price and perceived quality.

The findings suggest that Nigerian pharmaceutical firms should focus more on cost-efficiency and value communication rather than aggressive price cuts. In conclusion, the study provides empirical evidence that an optimal pricing mix—anchored on cost recovery and customer value—can significantly enhance the sales performance and sustainability of pharmaceutical firms operating in Nigeria's dynamic market environment.

Table 1a: Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.849[a]	.721	.703	.76604

a. Predictors: (Constant), Cost-Plus Pricing, Value-Based Pricing, Competitive Pricing, Penetration Pricing, Skimming Pricing

Table 1b. ANOVA[b]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	115.414	5	23.083	39.335	.000[a]
	Residual	44.598	76	.587		
	Total	160.012	81			

a. Predictors: (Constant), Cost-Plus Pricing, Value-Based Pricing, Competitive Pricing, Penetration Pricing, Skimming Pricing

b. Dependent Variable: Sales performance

Table 1c: Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	1.046	.633		1.652	.103
	Cost-Plus Pricing	.861	.082	.824	10.491	.000
	Value-Based Pricing	.118	.085	.118	1.384	.171
	Competitive Pricing	-.231	.104	-.172	-2.227	.029
	Penetration Pricing	-.158	.081	-.139	-1.960	.054
	Skimming Pricing	.072	.079	.058	.912	.364

a. Dependent Variable: Sales performance

Discussion of Findings

The regression results indicate that pricing strategy plays a crucial role in determining the sales performance of pharmaceutical firms in Nigeria, as evidenced by the high

coefficient of determination and the statistically significant F-value. This finding is consistent with Uzor, Okoye, and Okafor (2024), who reported that pricing behavior is one of the most critical determinants of sales outcomes in Nigeria's drug market, particularly due to inflationary pressures and rising import costs. Similarly, Adegoke and Ajibola (2022) found that pricing policies significantly influence competitive advantage and sales outcomes among Nigerian drug firms, emphasizing that pricing has become a strategic weapon rather than a mere operational decision. The strong explanatory power of this model therefore confirms that variations in pricing approaches directly shape firms' market performance and profitability within the country's competitive pharmaceutical sector.

The analysis further shows that Cost-Plus Pricing has a strong positive and statistically significant effect on sales performance. This suggests that Nigerian pharmaceutical firms benefit from pricing their products by adding a mark-up to production costs, ensuring stable revenue even amid fluctuating import and operational expenses. This result supports the conclusion of Oyelade and Ogunleye (2023), who observed that cost-based pricing strategies are particularly effective in highly regulated industries such as pharmaceuticals, where price controls and cost pressures dominate. Similarly, Okafor (2020) emphasized that cost-plus pricing helps firms maintain profitability under government price regulation, as it allows them to recover production expenses while sustaining operations. Hence, the present study aligns with prior research showing that cost-based approaches enhance both stability and sales growth in Nigeria's pharmaceutical industry.

On the other hand, Competitive Pricing was found to have a negative and significant effect on sales performance. This suggests that aggressive price competition, often used to undercut rivals, can erode profit margins and ultimately reduce overall firm performance. Bello and Oyetunde (2023) similarly reported that excessive price competition among drug firms led to short-term sales surges but long-term profit declines. In the same vein, Adegoke and Ajibola (2022) found that firms engaged in continuous price-matching strategies often lose their ability to invest in marketing and innovation, weakening their market sustainability. This evidence supports the notion that, while competitive pricing may help firms attract customers in the short run, it can be counterproductive in sustaining profitability and long-term performance in Nigeria's pharmaceutical market.

The study also shows that Penetration Pricing had a marginally negative impact on sales performance, while Value-Based Pricing and Skimming Pricing exhibited positive but statistically insignificant effects. These findings mirror those of Usman and Bello (2021), who found that value-based pricing in Nigeria's pharmaceutical sector improves customer trust and brand perception but requires strong product differentiation to significantly influence sales. Kumar and Sharma (2021) also reported that while value-based and skimming pricing positively affect revenue growth in

India's pharmaceutical industry, their success depends on consumer awareness and perceived product uniqueness. Furthermore, Aliyu and Ahmed (2019) argued that penetration pricing often leads to revenue losses because low introductory prices reduce perceived product quality and brand value. The insignificant impact of value-based and skimming pricing in this study therefore suggests that Nigerian pharmaceutical consumers remain largely price-sensitive, limiting the effectiveness of premium pricing strategies.

The findings confirm that pricing strategies significantly affect sales performance, with Cost-Plus Pricing emerging as the most effective approach in the Nigerian pharmaceutical context. This result reinforces the arguments of Oyelade and Ogunleye (2023) and Okafor (2020) that pricing decisions anchored on cost recovery enhance firm stability in turbulent economic conditions. However, the negative impacts of Competitive and Penetration Pricing strategies highlight the risks of relying solely on low-price tactics in a market characterized by high import dependence and regulated margins. As Uzor et al. (2024) cautioned, aggressive price cuts can undermine sustainability, especially when inflation and currency volatility constrain firms' cost structures.

Conclusion

The study concludes that pricing strategy is a key determinant of sales performance in the Nigerian pharmaceutical industry. Firms that adopt pricing models aligned with cost recovery, market conditions, and consumer purchasing power tend to perform better. However, strategies that focus solely on competition or temporary market penetration can be detrimental to long-term growth. The findings highlight the need for firms to strike a balance between affordability, profitability, and customer value in order to achieve sustainable performance in the face of economic volatility and regulatory constraints.

Recommendations

The following recommendations are proposed;

1. Pharmaceutical firms should continue to apply cost-plus pricing strategies to ensure that selling prices adequately cover production costs and desired profit margins. This approach will help maintain profitability despite rising operational expenses.
2. Firms should minimize reliance on price wars and instead focus on product differentiation through quality improvement, branding, and customer service. Competing on non-price factors will promote long-term stability and profitability.
3. Pharmaceutical firms should invest in creating customer awareness about the quality, safety, and effectiveness of their products. By highlighting product

value and benefits, firms can justify moderate price increases and strengthen brand loyalty.

4. Management should adopt modern pricing analytics and market intelligence tools to monitor market trends, consumer preferences, and competitors' pricing behavior. This will enable more informed and flexible pricing decisions.
5. The government should provide a stable pricing framework and incentives to local manufacturers to encourage affordable pricing and consistent drug supply. This will enhance both firm performance and public access to essential medicines.

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